

# NiFi : Prepend (orAppend) a line number to all records in a flowfile

Asked 1 month ago   Active 22 days ago   Viewed 39 times



I am using NiFi 1.9.2

0



I am reading a text file which happens to be a csv file. I have the Contents of the file in the Contents of a flowFile.



Contents are



```
a,b,c  
d,e,f  
g,h,i
```

I want to prepend a line number to all records in the flowfile and get

```
1,a,b,c  
2,d,e,f  
3,g,h,i
```

each time I feed a file through this processor

I can achieve something close by using the ReplaceText processor with Properties as follows:

```
Search Value : (?m)(^.*$)  
Replacement Value : ${nextInt()}, $1
```

But because nextInt() persists it's value over the lifetime of the running NiFi instance I get

```
0,a,b,c  
1,d,e,f  
2,g,h,i  
for 1st execution  
  
3,a,b,c  
4,d,e,f  
5,g,h,i  
for the next execution etc
```

Additionally, from the NiFi expression-language-guide, the "counter is shared across all NiFi components, so calling this function multiple times from one Processor will not guarantee sequential values within the context of a particular Processor."

1. Is there a way to ensure the line numbers always start at 0 for each execution of this processor for the lifetime of the NiFi instance, and are always sequential?
2. What the range of the counter?
3. Can I get the counter to start at 1?



You can try - `ExecuteScript` or `UpdateAttribute` to maintain a local state for counter variable, refer to this - [nifi.apache.org/docs/nifi-docs/components/org.apache.nifi/...](https://nifi.apache.org/docs/nifi-docs/components/org.apache.nifi/) – Vikramsinh Shinde Sep 10 at 8:52

Thanks @VikramsinhShinde. These processors are very useful for performing an operation on a flowFile-by-flowFile basis. I want to perform an operation on individual records in a flowFile. I can write a custom processor to this in Java quite easily, but first choice is to try a out-of-the-box NiFi solution. – mikec Sep 10 at 10:52

I have written a custom script to implement exactly this use case. I haven't come across any other solution. – Mallik Sep 17 at 15:25

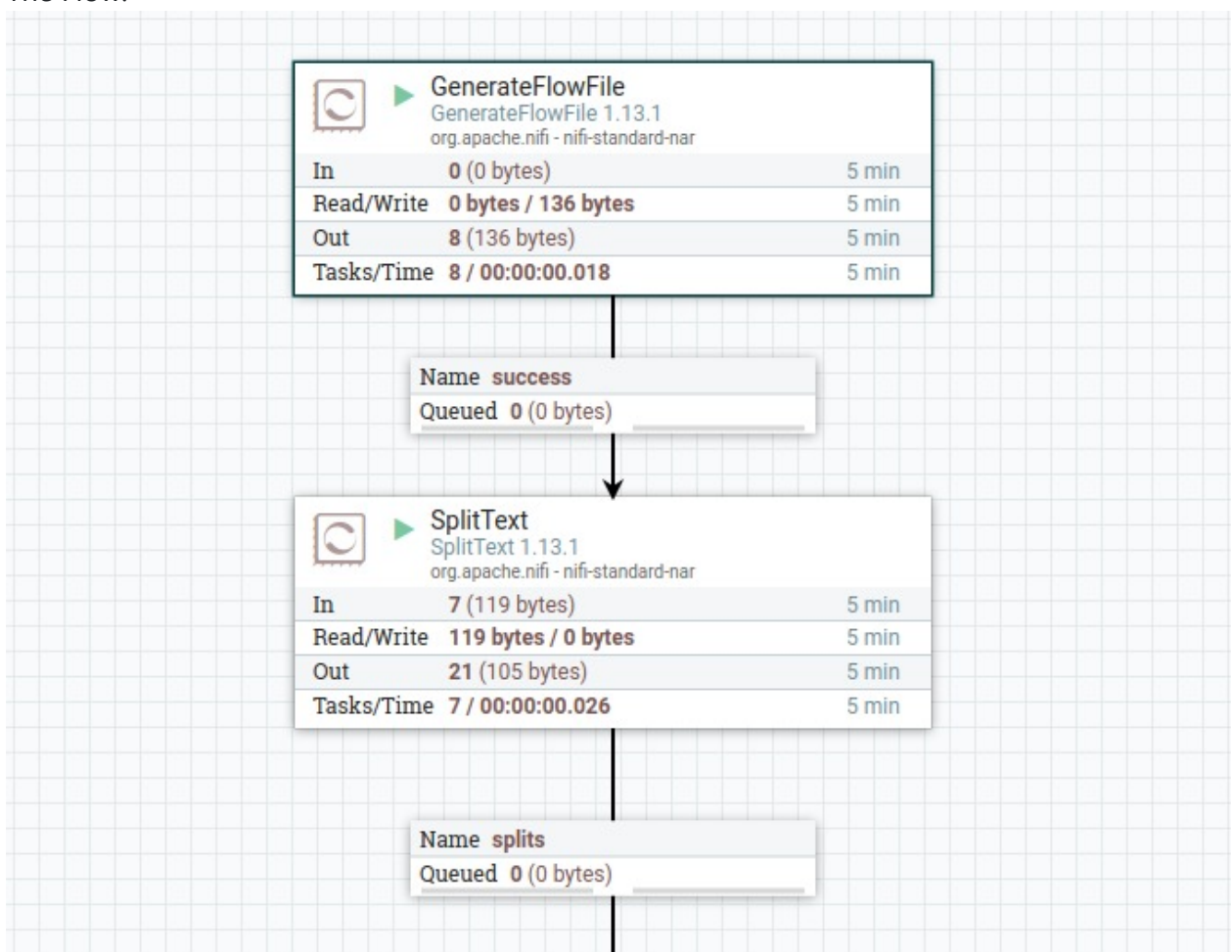
Thanks @Mallik. I did the same - wrote a custom processor in Java. – mikec Sep 21 at 8:52

## 1 Answer

Active Oldest Votes

You can split the content to several lines then use `fragment.index` to prepent the counter to the lines. After that you can merge them again.

The Flow:



 **ReplaceText**  
ReplaceText 1.13.1  
org.apache.nifi - nifi-standard-nar

|            |                       |       |
|------------|-----------------------|-------|
| In         | 22 (122 bytes)        | 5 min |
| Read/Write | 122 bytes / 170 bytes | 5 min |
| Out        | 22 (170 bytes)        | 5 min |
| Tasks/Time | 22 / 00:00:00.045     | 5 min |

Name **success**  
Queued 0 (0 bytes)

 **MergeContent**  
MergeContent 1.13.1  
org.apache.nifi - nifi-standard-nar

|            |                       |       |
|------------|-----------------------|-------|
| In         | 18 (126 bytes)        | 5 min |
| Read/Write | 126 bytes / 136 bytes | 5 min |
| Out        | 6 (136 bytes)         | 5 min |
| Tasks/Time | 95 / 00:00:00.059     | 5 min |

Name **merged**  
Queued 4 (92 bytes)



## GenerateFlowFile:

### Processor Details

▶ Running

⚙ STOP & CONFIGURE

SETTINGS

SCHEDULING

PROPERTIES

COMMENTS

Required field

| Property         |   | Value             |
|------------------|---|-------------------|
| File Size        | ? | 0B                |
| Batch Size       | ? | 1                 |
| Data Format      | ? | Text              |
| Unique FlowFiles | ? | false             |
| Custom Text      | ? | a,b,c d,e,f g,h,i |
| Character Set    | ? | UTF-8             |
| Mime Type        | ? | No value set      |

OK

## SplitText:

### Processor Details

▶ Running

⚙ STOP & CONFIGURE

SETTINGS

SCHEDULING

PROPERTIES

COMMENTS

Required field

| Property                      |   | Value        |
|-------------------------------|---|--------------|
| Line Split Count              | ? | 1            |
| Maximum Fragment Size         | ? | No value set |
| Header Line Count             | ? | 0            |
| Header Line Marker Characters | ? | No value set |
| Remove Trailing Newlines      | ? | true         |

OK

ReplaceText:

### Processor Details

▶ Running

⚙ STOP & CONFIGURE

SETTINGS

SCHEDULING

PROPERTIES

COMMENTS

Required field

| Property                     |   | Value               |
|------------------------------|---|---------------------|
| Search Value                 | ? | (?s)(^)             |
| Replacement Value            | ? | \${fragment.index}, |
| Character Set                | ? | UTF-8               |
| Maximum Buffer Size          | ? | 1 MB                |
| Replacement Strategy         | ? | Prepend             |
| Evaluation Mode              | ? | Line-by-Line        |
| Line-by-Line Evaluation Mode | ? | All                 |

OK

MergeContent:

### Processor Details

▶ Running

⚙ STOP & CONFIGURE

SETTINGS

SCHEDULING

PROPERTIES

COMMENTS

Required field

| Property               |   | Value                       |
|------------------------|---|-----------------------------|
| Merge Strategy         | ? | Defragment                  |
| Merge Format           | ? | Binary Concatenation        |
| Attribute Strategy     | ? | Keep Only Common Attributes |
| Max Bin Age            | ? | No value set                |
| Maximum number of Bins | ? | 5                           |
| Delimiter Strategy     | ? | Text                        |
| Header                 | ? | No value set                |
| Footer                 | ? | No value set                |
| Demarcator             | ? |                             |

OK

Don't forget to add a new line (Shift+Enter) to Demarcator attribute.



View as: original



|   |         |
|---|---------|
| 1 | 1,a,b,c |
| 2 | 2,a,b,c |
| 3 | 3,a,b,c |

Result:

You can use `${Fragment.index:minus(1)}` if you want to count from zero.

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edited Sep 21 at 15:02

answered Sep 17 at 15:45



**Behrouz Seyedi**

**300** 1 14

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Thank you @Behrouz. That's pretty smart. In SplitText I had to change the 'Remove Trailing Newlines' to false in order to get the merged output to create each record on a new line. Question : SplitText creates a new flowFile for each record of the input file. MergeContent recombines them. From the tests I ran it appears that the original order of records is preserved. But is this guaranteed? – [mikec](#) Sep 21 at 8:51

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@mikec As I know the order of records will be guaranteed to preserve. In MergeContent, "fragment.index" attribute indicates the order in which the fragments should be assembled.  
– [Behrouz Seyedi](#) Sep 21 at 14:58

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@mikec I had added new line for each line in "Demarcator" attribute of MergeContent.  
– [Behrouz Seyedi](#) Sep 21 at 15:00

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